



Exploring sex differences in human health risk assessment for PFNA and PFDA using a PBPK model

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Abstract

Perfluorononanoic acid (PFNA) and perfluorodecanoic acid (PFDA), which are classified as perfluoroalkyl and polyfluoroalkyl substances (PFASs), have been widely used in industrial applications as a surface protectant. PFASs have been detected in wildlife and in humans around the globe. The purposes of this study are to develop and validate a physiologically based pharmacokinetic (PBPK) model for detecting PFNA and PFDA in male and female rats, and to apply the model to a human health risk assessment regarding the sex difference. A PBPK model of PFNA and PFDA was established based on an in vivo study in male and female rats. Analytes in biological samples (plasma, nine tissues, urine, and feces) were determined by ultra-liquid chromatography coupled tandem mass spectrometry (UPLC–MS/MS) method. PFNA and PFDA showed a gender differences in the elimination half-life and volume of distribution. The tissue–plasma partition coefficients were the highest in the liver in both male and female rats. The predicted rat plasma and urine concentrations simulated and fitted were in good agreement with the observed values. The PBPK models of PFNA and PFDA in male and female rats were then extrapolated to a human PBPK model based on human physiological parameters. The external doses were calculated at 3.35 ng/kg/day (male) and 17.0 ng/kg/day (female) for PFNA and 0.530 ng/kg/day (male) and 0.661 ng/kg/day (female) for PFDA. Human risk assessment was estimated using Korean biomonitoring values considering the gender differences. This study provides valuable insight into human health risk assessment regarding PFNA and PFDA exposure.

Keywords PBPK · Perfluorononanoic acid (PFNA) · Perfluorodecanoic acid (PFDA) · Human health risk assessment · Rat

Abbreviations

AUC_{inf}	Area under the concentration–time curve from zero to infinity
CL	Clearance
C_{max}	Maximum plasma concentration
C_{ss}	Steady-state concentration
F_r	Free fraction in plasma
IV	Intravenous
K_p	Tissue-to-plasma partition coefficient
K_{st}	Rate constant to storage compartment
K_t	Transporter affinity constant
K_u	Urinary elimination rate constant
MOE	Margin of exposure

NOAEL	No-observed-adverse-effect level
PBPK	Physiologically based pharmacokinetic model
PFASs	Perfluoroalkyl and polyfluoroalkyl substances
PFHxS	Perfluorohexanesulfonic acid
PFDA	Perfluorodecanoic acid
PFNA	Perfluorononanoic acid
PFOA	Perfluorooctanoic acid
PFOS	Perfluorooctanesulfonic acid
PK	Pharmacokinetic
POD	Point of departure
$t_{1/2}$	The elimination half-life
TDI	The tolerable daily intake
T_m	Transport maximum
UF	Uncertainty factor
UF_A	Uncertainty factor for interspecies extrapolation from rats to humans
UF_H	Uncertainty factor for inter-individual variability in humans
UF_S	Uncertainty factor for subchronic to chronic extrapolation
V_d	Volume of distribution

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Introduction

Perfluoroalkyl and polyfluoroalkyl substances (PFASs) have been widely used in industrial applications for surface protectant in paper cardboard packaging products, carpets, leather products, and textiles (ATSDR 2018a). Since the PFASs have properties of high stability, persistence, bioaccumulation, and toxicity, they have been found globally in humans and the environment, and are an increasing public health concern (Bartolome et al. 2017). Among the PFASs, perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS) are the most well-known (Kim et al. 2016b), followed by perfluorohexane sulfonic acid (PFHxS), perfluorononanoic acid (PFNA), and perfluorodecanoic acid (PFDA) in sequence. While many studies have been performed on PFOA, PFOS, and PFHxS, information on PFNA and PFDA is limited. Therefore, the present study focused on PFNA and PFDA.

Among the widely detected PFASs, the toxicity by the exposure of PFNA and PFDA has continued to be reported. Serum PFNA concentration in prenatal and early childhood has a positive association with children's reading scores at the age range of 5–8 years (Zhang et al. 2018). PFDA exposed during prenatal development significantly increased the risk of childhood atopic dermatitis (AD) in female children during the first 24 months of life, and PFNA showed the statistical significance between childhood AD and prenatal exposure (Chen et al. 2018). Women, who used oral contraceptives for more than 10 years, showed increased PFNA plasma concentrations compared with women who had never used oral contraceptives (Rush et al. 2018). Exposure of PFNA was shown to cause the hepatocytes, hepatic lipid metabolism and lipid accumulation in the liver (Fang et al. 2012b). In addition, PFNA disturbs glucose metabolism through the inhibition of the hepatic insulin signal pathway and accelerates the output of glucose and increasing glycogen synthesis in the rat liver (Fang et al. 2012a). PFNA exposure showed a significant negative relationship with bone parameter and was associated with raised systolic blood pressure, low density lipoprotein cholesterol, and total cholesterol (Khalil et al. 2018). PFNA and PFDA potentially induce the activity of peroxisomal β -oxidation in the liver of male and female rats (Kudo et al. 2000). Kupffer cells have been reported to be involved in PFDA-mediated transcriptional changes in hepatocytes via either inflammatory or oxidative mediators (Maher et al. 2008).

In Korea, Seo et al. (2018) reported that the serum concentration of the 13 PFASs in adults living in Seoul was gradually increased between 2006 and 2013, and then decreased. The PFNA and PFDA concentrations in males (2.23 ± 1.80 ng/mL, PFNA; 1.47 ± 1.30 ng/mL, PFDA)

were higher than in females (1.96 ± 1.72 ng/mL, PFNA; 1.22 ± 1.07 ng/mL, PFDA) (Seo et al. 2018). The PFDA showed significant ($p < 0.05$) difference (Seo et al. 2018). In the general population of Siheung, Korea, PFNA concentration showed a significant ($p < 0.01$) gender difference (male = median, 2.16 ng/mL and female 2.02 ng/mL), but PFDA did not show a significant difference (male 0.98 ng/mL and female 0.86 ng/mL) (Ji et al. 2012). The concentrations of PFNA and PFDA were 19.4 ± 21.8 ng/L (63% of samples) and 0.88 ± 5.57 ng/L (9% of samples), respectively, in breast milk obtained from mothers in Korea (Lee et al. 2018) and 1.931 ng/mL and 0.523 ng/mL in infant serum (Kim et al. 2016a), respectively. The Agency for Toxic Substances and Disease Registry (ATSDR) measured the concentrations of the PFASs in blood and urine from local residents to estimate PFASs exposure over time. The PFNA level was significantly decreased in 2016 compared to 2010 (–58%) (Worley et al. 2017). PFNA and PFDA were detected in 99.9% (0.96 ng/mL) and 86.4% (0.42 ng/mL), respectively, of serum samples obtained from Spanish adults (age range of 18–65 years). The PFNA showed a significant ($p < 0.05$) gender difference (PFNA concentration in men was higher than in women), while PFDA did not show a significant difference between male and female (Bartolome et al. 2017).

For the risk assessment of PFNA and PFDA, which are known to cause the toxicity, physiologically based pharmacokinetic (PBPK) modeling is used as a mathematical tool for predicting the absorption, distribution, metabolism, and excretion of PFNA and PFDA. Therefore, a pharmacokinetic (PK) study of PFNA and PFDA first needs to be carried out for the development of the PBPK model. Several PKs of PFNA and PFDA have recently been reported. In rodents, the elimination half-life ($t_{1/2}$) of PFNA was estimated at 30.6 days (male) and 1.4 days (female) in Sprague–Dawley rats (Tatum-Gibbs et al. 2011) and at 29.5 days (male) and 2.4 days (female) in Wistar rats (Ohmori et al. 2003). The elimination $t_{1/2}$ of PFDA was reported at 39.9 days (male) and 58.6 days (female) in Wistar rats (Ohmori et al. 2003). In rats, PFNA and PFDA showed a significant gender difference in $t_{1/2}$. CD-1 mice given at single doses of 1 and 10 mg/kg of PFNA showed a $t_{1/2}$ of 34.3–68.9 days in male and 25.8–68.4 days in female (Tatum-Gibbs et al. 2011). While rats showed a gender difference in the elimination $t_{1/2}$ of PFNA, mice showed no difference. The PFNA was highly distributed in rats and mice liver (Tatum-Gibbs et al. 2011). In female microminipigs, the $t_{1/2}$ was estimated at 49.5 days for PFNA and 40.8 days for PFDA (Guruge et al. 2016). PFNA and PFDA were detected at the highest level in rat liver, and followed by kidney (Benskin et al. 2009; Guruge et al. 2016; Iwabuchi et al. 2017; Vanden Heuvel et al. 1991). When PFNA and PFDA were intraperitoneally injected into male and female rats, in male rats, they were

eliminated in the urine in the male rats at the proportions of 2.0% (PFNA) and 0.2% (PFDA) of the dose within 120 h (Kudo et al. 2001). PFNA in female rats was eliminated in the urine at the proportion of 51% of the dose within 120 h, significantly higher than that in male rats (Kudo et al. 2001). However, Vanden Heuvel et al. (1991) reported that PFDA-derived ^{14}C showed a gender difference, with 51% (male) and 24% (female) of administered PFDA in fecal elimination (Vanden Heuvel et al. 1991). This report was not consistent that by Kudo et al. (2001). Therefore, the PKs of PFDA should be clearly characterized to develop the PBPK model.

The PBPK models of PFASs have been continued to be reported. The PBPK models of PFOA and PFOS have been developed for monkeys (Loccisano et al. 2011) and rats (Loccisano et al. 2012a, b). One PBPK model was described the relationship between serum concentration of PFOA and PFOS and the early onset of menopause (Ruark et al. 2017), or delayed menarche (Wu et al. 2015). The prenatal and postnatal exposure of PFHxS, PFOA, and PFOS was estimated using the PBPK model (Verner et al. 2016). In our recent report, the PBPK model of PFHxS was developed and applied to the human health risk assessment of PFHxS on the gender differences (Kim et al. 2018). To the best of the authors' knowledge, a PBPK model of PFNA and PFDA has not been developed and the risk assessment considering the gender difference of PFNA and PFDA was not estimated. This study might therefore be the first report of a PBPK model of PFNA and PFDA for gender-specific risk assessment.

Several reports have examined in health risk assessment of PFASs. These studies reported the hazard quotients (HQs) of PFOA and PFOS in parental and cord serum (Han et al. 2012), in the cord blood of newborns (Wang et al. 2016), in the atmosphere of Shenzhen in China (Liu et al. 2015) and in the Swedish population (Borg et al. 2013). Although more than 13 PFASs were examined in these studies, HQ was calculated only for the highest detected the PFASs such as PFOA and PFOS. In 2018, the ATSDR (draft, June, 2018) reported on the minimal risk levels of PFNA (0.003 $\mu\text{g}/\text{kg}/\text{day}$) for toxicity development at endpoint after oral administration, whereas PFDA has not found in this report (ATSDR 2018b). The health risk assessment of PFNA and PFDA has thus far not been reported. The World Health Organization (WHO 2010) and Organisation for Economic Co-operation and Development (OECD 2015) have suggested using the PBPK model as a guide for risk assessment.

The purposes of this study were to develop and validate a PBPK model of PFNA and PFDA in male and female rats, and to evaluate the gender-specific risk assessment for human exposure using the PBPK model. The developed PBPK model in male and female rats was extrapolated to the human model, and simulated to estimate the risk assessment of PFNA and PFDA. This study provides valuable insight

into the human health risk assessment of PFNA and PFDA exposure and can be used to compare the risk assessments of male and female humans.

Materials and methods

Materials and reagents

The reference standard of PFNA (purity > 99.5%) and PFDA (purity > 99.5%) was obtained from Sigma-Aldrich (St. Louis, MO, USA). Perfluoro-[1,2,3,4,5- $^{13}\text{C}_5$] nonanoic acid (MPFNA, purity > 98%), and Perfluoro-n-[1,2- $^{13}\text{C}_2$] deca-noic acid (MPFDA, purity > 98%) were supplied by Wellington Laboratories Inc. (Guelph, ON, Canada) and used as an internal standard (IS). The other chemicals and reagents used for this study were of HPLC grade. Distilled water was supplied from an Elga Purelab Option-Q system (Elga Lab-Water, Marlow, UK).

Animals

Female and male Sprague–Dawley rats (7–8 weeks, 200–220 g) were purchased from Damul Sciences (Daejeon, Republic of Korea). All animals were individually housed in a metabolic cage that controlled to maintain a temperature of $23 \pm 2^\circ\text{C}$, a relative humidity of $50 \pm 5\%$, and a 12-h light–dark cycle. This study was performed at CHA Laboratory Animal Research Center and approved by the Institutional Animal Care and Use Committee (IACUC, No. IACUC150029 and IACUC160008).

Pharmacokinetic study in rats

The PK study consisted of minimum experiments designed to develop the PBPK model considering sex, administration routs, and doses. For the PFNA study, male rats were divided into two groups and were given a dose of 3 mg/kg PFNA by oral or intravenous (IV) administration ($n = 5/\text{group}$). In the female rats, PFNA was administered by orally (at doses of 0.5, 1, 3, and 10 mg/kg) to estimate the dose-dependent linearity and by IV injection (at doses of 0.5, 1, and 3 mg/kg) to obtain the partition coefficient (K_p) values ($n = 5/\text{group}$). Blood samples were collected from the jugular vein at 0, 0.25, 0.5, 1, 2, 4, 8, 12, and 24 (h), and at 3, 5, 10, 20, 30, 50, and 60 days for male rats and at 0, 0.25, 0.5, 1, 2, 4, 8, 12, and 24 (h), and at 2, 3, 5, 10 days and every 10 days after the administration during the study period (until 30 days at a dose of 0.5 mg/kg, until 40 days at a dose of 1 mg/kg, and until 60 days at a dose of 3 and 10 mg/kg) for female rats. In the PFDA study, male and female rats were both divided into two groups and given a dose of 1 mg/kg PFDA via oral or IV administration ($n = 5/\text{group}$). Blood samples were

collected at 0, 0.25, 0.5, 1, 2, 4, 8, 12, and 24 (h), and at 3, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, and 150 days.

After taking the blood samples, they were immediately heparinized and centrifuged at 7000 rpm for 10 min. Plasma obtained from the blood samples was kept at -80°C until the assay. All rats were sacrificed after taking the blood sample at a terminal time. The brain, heart, liver, lung, spleen, kidney, gastrointestinal (GI) tract, adipose tissue, and muscle were collected and stored in a freezer at -80°C until the analysis. The metabolic cages were used to collect the rat urine and feces. After collection, the total mass of feces and total volume of urine were measured. All samples were stored at -80°C until assay. The PK parameters were estimated using WinNonlin[®] software (version 8.0, Pharsight[®], Certara[™] Company, Princeton, NJ, USA).

Determination of PFNA and PFDA in rat biological samples

An Acquity UPLC[®] system coupled with a mass spectrometer (Xevo TQ-S, Waters Corp., Milford, MA, USA) was used to determine the PFNA and PFDA concentrations in rat biological samples. Chromatography separation was carried out using a Halo C₁₈ column (2.1 mm × 50 mm, 2.7 μm particle size) at a temperature of $40 \pm 0.5^{\circ}\text{C}$. The mobile phase was composed of 2 mM ammonium acetate in water and acetonitrile at a flow rate of 0.2 mL/min of flow rate. The gradient conditions for the mobile phase were as follows: at 0–0.5 min, 30% of B; at 0.5–2.0 min, 30–80% of B; at 2.0–2.1 min, 80–90% of B; at 2.1–3.4 min, 90% of B; at 3.5–4.5 min, 30% of B. The mass spectrometer was operated with an electrospray ionization interface in the negative ion mode. Multiple reaction monitoring (MRM) transitions were m/z 463.0 → 419.0 for PFNA, m/z 468.0 → 423.1 for MPFNA (IS of PFNA), m/z 513.0 → 469.0 for PFDA, and m/z 515.1 → 470.0 for MPFDA (IS of PFDA). The mass parameters of PFNA and PFDA were optimized at 2.8 kV for the capillary voltage, 10 V for cone voltage, and 10 eV for collision energy. Data acquisition and analysis were carried out using the Masslynx 4.1 software (Waters Corp., Milford, MA, USA).

PFNA has a standard calibration range of 0.5–1000 ng/mL in rat plasma, tissues, urine, and feces. The calibration ranges of PFDA were 0.5–1000 ng/mL in plasma and 0.5–500 ng/mL in tissues, urine, and feces. PFNA and PFDA in the biological samples were extracted using a combination of protein precipitation and liquid–liquid extraction. The plasma samples were diluted with distilled water (1:4) before the extraction. All tissues and feces were added with distilled water (five times volume of sample amount) and homogenized (Ultra-Turrax[®], IKA[®] T10 basic). Then, each 0.2 mL sample was added to 0.01 mL of the IS solution (100 ng/

mL in 50% methanol) and 0.6 mL acetonitrile. After vortex-mixing for 1 min and centrifuging at $15,000\times g$ for 5 min, 0.6 mL of the supernatant was transferred to a new tube. 0.6 mL of methylene chloride was then added, then vortex-mixed for 3 min and then centrifuged at $15,000\times g$ for 5 min. An organic layer of 0.6 mL was transferred into a clean tube and dried under a gentle stream of nitrogen gas at 50°C . The residues were reconstituted with 30% acetonitrile and 2 μL of the supernatant was injected into UPLC–MS/MS. The samples exceeding the calibration range were reanalyzed after dilution. The developed analytical method for PFNA and PFDA was validated in accordance with the Guidance for Industry: Bioanalytical Method Validation (FDA 2013).

Protein binding assay in rat and human plasma

Protein binding assay was investigated in male and female rats and human plasma with using the Centrifree[®] Ultrafiltration Device (Merck Millipore Ltd, Darmstadt, Germany). The procedure was conducted according to our previous study (Kim et al. 2018) and the user guide of this device. The PFNA and PFDA of 1 μg/mL concentration were evaluated and measured by the developed UPLC–MS/MS. The percentage of plasma protein binding (1) was calculated using Eq. (1) as follows, where C_{UF} and C_{total} refer to the concentration in the ultrafiltrate (C_{UF}) and total concentration in plasma, respectively,

$$\text{Protein binding (\%)} = (C_{\text{total}} - C_{\text{UF}}) / C_{\text{total}} \times 100 \quad (1)$$

Tissue-to-plasma partition coefficients

The tissue-to-plasma partition coefficients (K_p) of PFNA and PFDA in male and female rats were obtained from an in vivo study. For each organ, the K_p values were calculated as the ratio of plasma and organ concentration at terminal time after IV injection in rats. The following equation was used to obtain the K_p in each organ (Fig. 1).

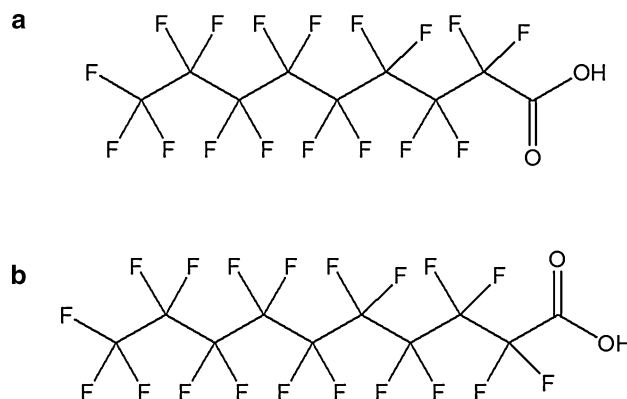


Fig. 1 Chemical structures of **a** PFNA and **b** PFDA

$$K_p = \frac{C_{\text{tissue}}}{C_{\text{plasma}}} \quad (2)$$

Development of PBPK model

We previously reported that the PBPK model of PFHxS was developed and applied to human health risk assessment after exposure of PFHxS (Kim et al. 2018). In the present study, we adapted the methodology of the previous report and modified it appropriately for PFNA and PFDA. The PK and pharmacodynamic properties of the general PFASs were considered to develop a PBPK model for PFNA and PFDA. The model compartments consisted of blood, filtrate, storage, and tissues interconnected via blood flow. The PBPK model of PFDA contained the one more storage compartment which is located on the path from liver to feces, regarding the major excretion route of PFDA. PFDA was known to mainly eliminate via feces (Kudo et al. 2001). The nine tissues (liver, kidney, lung, heart, brain, adipose tissue, GI tract, muscle, and spleen) were considered in the PBPK model. The tissue compartments comprised of the GI tract (absorption site), liver (accumulative site), kidney (a renal reabsorption by filtration compartment), and other tissues that were considered as the remaining level after PFNA and PFDA exposure. The menstrual blood loss was also considered as one of the excretion routes for developing the PBPK

model for human females. The structures of the PBPK model for PFNA and PFDA in rats are shown in Fig. 2.

The physiological and biochemical parameters were used to develop the rat PBPK model of PFNA and PFDA and to extrapolate a human model. The physiological parameters are referred from a previous study (Kim et al. 2018) and are represented in Table 1. The biochemical parameters used to explain the model of PFNA and PFDA are as follows: the transport maximum (T_m), the transporter affinity constant (K_t), the rate constant to storage compartment (K_{st}), the free fraction in plasma (F_f), and the urinary elimination rate constant (K_u), the fecal elimination constant (K_f), and the biliary excretion rate to storage (K_{bile}). The values of T_m , K_t , K_{st} , K_u , K_f , and K_{bile} were estimated by fitting the plasma, urine or feces concentration from the in vivo study, whereas the F_f of PFNA and PFDA in plasma was obtained from a protein binding assay. The filtrate compartment was included in this model to describe the renal reabsorption of PFNA and PFDA. This model was similar to the PBPK model of PFHxS in rats (Kim et al. 2018) and PFOA and PFOS in monkeys (Loccisano et al. 2011).

Considering the exposure routes of PFNA and PFDA, differential mass balance equations were derived by oral administration. The equations were used to determine the PFNA and PFDA level in the plasma, in each tissue connected to the plasma, the oral compartment, the liver and kidney compartments, the filtrate, and the storage compartment after

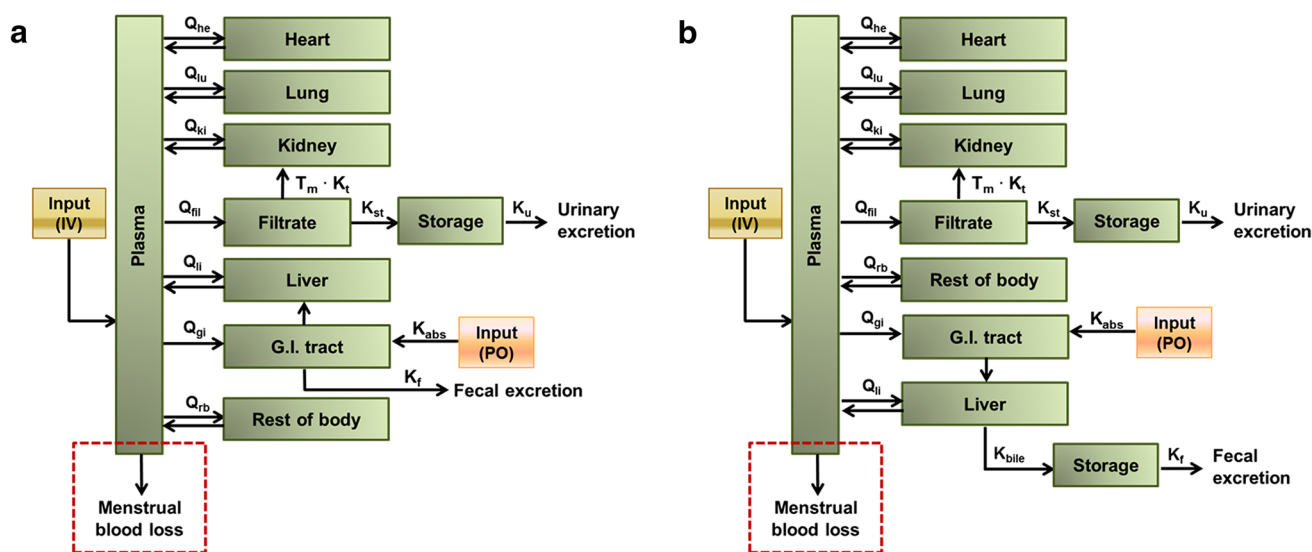


Fig. 2 Structures of the PBPK model for **a** PFNA and **b** PFDA in rats and humans. Both models reflect the renal reabsorption of PFASs to kidney, and PFDA model was added to the storage compartment on the path from liver to feces, considering the major excretion route of PFDA. The transport maximum (T_m) and transporter affinity constant (K_t) were a key parameter for reabsorption of PFASs to the kidney. The model included the other rate constants which are a rate constant to the storage compartment (K_{st}), the urinary elimination rate con-

stant (K_u), the transfer rate constant to fecal elimination (K_f), biliary excretion rate to storage (K_{bile}), and the oral absorption rate constant (K_{abs}). The Q_{he} , Q_{lu} , Q_{ki} , Q_{li} , Q_{gi} , and Q_{rb} mean the blood flow rates of the tissue connected to the blood such as heart, lung, kidney, liver, GI tract, and rest of body, respectively. The Q_{fil} indicates a clearance from the plasma to the filtrate compartment and the red square indicates menstrual blood loss in female humans

Table 1 Physiological parameters for the PBPK model

Tissue	Rat		Human	
	Tissue volume (mL)	Blood flow (mL/h)	Tissue volume (mL)	Blood flow (mL/h)
Blood	20.4	2580	5400	379,800
Heart	0.8	234	300	14,400
Brain	0.7	18	1500	42,000
Lung	1.0	2580	1200	379,800
Adipose tissue	10.0	24	12,200	15,600
Muscle	122.0	450	30,000	45,000
Kidney	1.8	520	300	74,400
Filtrate	0.18	260	30	14,880
G.I tract	10.0	450	2400	66,000
Liver	8.0	828	1500	87,000
Spleen	1.3	36	192	4620

Physiological parameters used in the PBPK model were referred to Kim et al. (2018)

oral administration. For the PBPK model fitting and simulation, the Berkeley Madonna V 8.3.14 software (Berkeley, CA, USA) was employed, which is a tool used for PBPK modeling in the risk assessment, as suggested by the United States Environmental Protection Agency (EPA 2006).

Model validation

The developed PBPK model was validated by comparing the simulated concentration and experimental data of the PFASs obtained from the in vivo study. The PBPK model should be able to describe the PK data of PFNA and PFDA in male and female rats. The observed values obtained from biological samples (plasma, tissues, urine, and feces) were visually compared to the predicted concentration by model simulation. The experimental data and the predicted values should be in good agreement. To validate the PBPK model, the male and female rats were orally administered a dose of 0.5–3 mg/kg for PFNA ($n=5/\text{group}$) and a dose of 1 mg/kg of PFDA ($n=5/\text{group}$), considering the exposure route of PFASs.

For the developed PBPK model of PFNA and PFDA, a normalized sensitivity analysis was carried out by the area under the concentration–time curve (*AUC*) obtained from the original parameter and the changed parameter after increasing by 1%. Each biochemical parameter was individually increased by 1%. The equation for calculating the normalized sensitivity coefficients (3) was as follows:

$$\text{Sensitive coefficient} = \frac{A - B}{B} \bigg/ \frac{C - D}{D} \quad (3)$$

where *A* is the *AUC* obtained from the 1% increase in the original parameter value, *B* is the *AUC* obtained from the

original parameter value, *C* is the parameter value increased by 1%, and *D* is the original parameter value. Plasma predictions for *AUC* were performed at 1 mg/kg dose of PFNA and PFDA after oral administration to male and female rats.

Extrapolation to PBPK model in humans

The PBPK model structure for PFNA and PFDA in humans was consistent with that in rats. The rat PBPK model was extrapolated to humans, considering the interspecies differences of physiological and chemical-specific parameters. The physiological parameters (tissue volume and blood flow) in rats were replaced with the corresponding physiological parameters for humans (Table 1). The average body weight (BW) of rats and humans was assumed to be 0.25 and 70 kg (58 kg for female humans), respectively, and all parameters of the same tissue for rats and humans were assumed to be identical. We estimated k_{human} using the following allometric equation (Kim et al. 2018):

$$k_{\text{human}} = k_{\text{rat}} \times \left(\frac{\text{BW}_{\text{human}}}{\text{BW}_{\text{rat}}} \right)^{-0.25} \quad (4)$$

where k_{human} and k_{rat} are the rate constants in humans and rats, respectively, BW_{human} and BW_{rat} refer to the body weight in humans and rats, respectively, and -0.25 is the allometric exponent. The T_m of humans was scaled as $\text{BW}^{0.75}$. The simulation of the PBPK model in male and female rats was performed at a no-observed-adverse-effect level (NOAEL) dose of 0.2 and 1.2 mg/kg, respectively, as a repeated administration (Fang et al. 2012b).

Human health risk assessment

Considering the exposure route of PFASs, we conducted the risk assessment using the points of departure (POD) based on the NOAELs after oral administration to rats. The POD was chosen as the point of most sensitivity to the effect of two PFASs exposure to rats. For the PFNA, the POD was selected for hepatic toxicity after exposure of PFNA (0.2 mg/kg/day) in the male rats (Fang et al. 2012b). The other NOAELs of PFNA were 5 mg/kg/day for endocrine and reproductive toxicity, 3–5 mg/kg/day for genetic toxicity (Fang et al. 2010), 1–5 mg/kg/day for apoptosis including factor increase, and 3–5 mg/kg/day for spleen toxicity (Fang et al. 2010). The POD of PFDA was chosen at 1.2 mg/kg/day for toxic hepatopathy in male rats (Kawashima et al. 1995) and the other NOAEL of PFDA was 3 mg/kg/day for weight loss in female mice (Harris et al. 1989). The risk assessment procedures of PFNA and PFDA were individually performed in the same manner.

The steady-state concentration (C_{ss}) of each PFAS in rats was calculated by entering the POD of PFAS to the rat

PBPK model. Based on the C_{ss} in rats, the C_{ss} in humans was predicted using an uncertainty factor (UF) with inter-species extrapolation. The UF used for extrapolation from rats to humans (UF_A) was $10^{0.5}$ (WHO 2010). According to the dose-metric approach (WHO 2010), we estimated the PODs in the male and female humans. According to the exposure scenario of PFAS repeated every 24 h, the external dose of PFAS was calculated using the human PBPK model plasma. The mean plasma concentration obtained from the repeated exposure scenario of PFAS was used to calculate the external dose of PFNA using the human PBPK model. The tolerable daily intake (TDI) was estimated by dividing the POD by UF, based on the guidance for risk assessment using the PBPK model by the US EPA (2006) and WHO (2010). The equation of TDI (5) was as follows:

$$TDI = POD/UF \quad (5)$$

where POD refer to a NOAEL in humans, and the UF was set at 10 for inter-individual variability in humans (UF_H) (WHO 2010) and 10 for subchronic to chronic extrapolation (UF_S) (Dourson and Stara 1983). We estimated the external dose using the PFAS concentration in humans, obtained from the human biomonitoring values of hazardous substances (including the PFNA and PFDA) by the Ministry of Food and Drug Safety (MFDS) in 2008 (MFDS 2009). The margin of exposure (MOE) indicates the margin between the reference dose and the actual exposure. It was calculated from the ratio of the TDI and the external exposure in humans. The equation of MOE (6) was as follows:

$$\text{Margin of exposure} = \frac{\text{Reference dose}}{\text{External exposure}} \quad (6)$$

Statistical analysis

The data were analyzed using the Statistical Analysis System (SAS) software version 9 (SAS Institute). Student's t test was used for statistical analysis and $p < 0.05$ indicates a significant difference.

Results

Pharmacokinetics of PFNA and PFDA

The PK parameters of PFNA were estimated using a two-compartment model after oral and IV administration of PFNA at doses of 0.5–10 mg/kg for female rats (10 mg/kg only for oral administration) and at doses of 3 mg/kg for male rats. Table 2 summarizes the PK parameters of PFNA after administration at a dose of 3 mg/kg and the mean plasma concentration–time profiles are shown in Fig. 3a, b. In the female rats, AUC_{inf} increased in a dose-dependency

Table 2 Pharmacokinetic parameters of PFNA and PFDA following oral or IV administration in male and female rats ($n = 5$, mean $\pm$ SD)

Parameters	PFNA (3 mg/kg)		PFDA (1 mg/kg)	
	Female		Male	
	IV	Oral	IV	Oral
AUC_{inf} ($\mu\text{g}\cdot\text{day}/\text{mL}$)	45.31 $\pm$ 3.51	52.88 $\pm$ 12.45	411.40 $\pm$ 79.67	313.64 $\pm$ 36.92
C_{max} ($\mu\text{g}/\text{mL}$) ^a	16.36 $\pm$ 1.30	12.80 $\pm$ 1.48	10.64 $\pm$ 5.85	8.19 $\pm$ 0.70
$t_{\alpha, 1/2}$ (day)	1.22 $\pm$ 0.27	1.64 $\pm$ 0.92	1.65 $\pm$ 1.19	3.00 $\pm$ 0.93
$t_{\beta, 1/2}$ (day) ^b	4.44 $\pm$ 0.16	6.40 $\pm$ 1.11	40.20 $\pm$ 18.68*	54.57 $\pm$ 2.54
V_d (mL)	45.86 $\pm$ 3.65	–	363.09 $\pm$ 182.69*	–
CL (mL/day)	16.55 $\pm$ 1.28	–	7.43 $\pm$ 1.44*	–
F	NA	0.79 $\pm$ 0.15	NA	0.77 $\pm$ 0.06

NA not available

* $p < 0.05$ between male and female

^aThe concentration is C_0 ($\mu\text{g}/\text{mL}$) after IV administration

^bThe value of PFDA indicates the elimination $t_{1/2}$ by estimating the one-compartment model

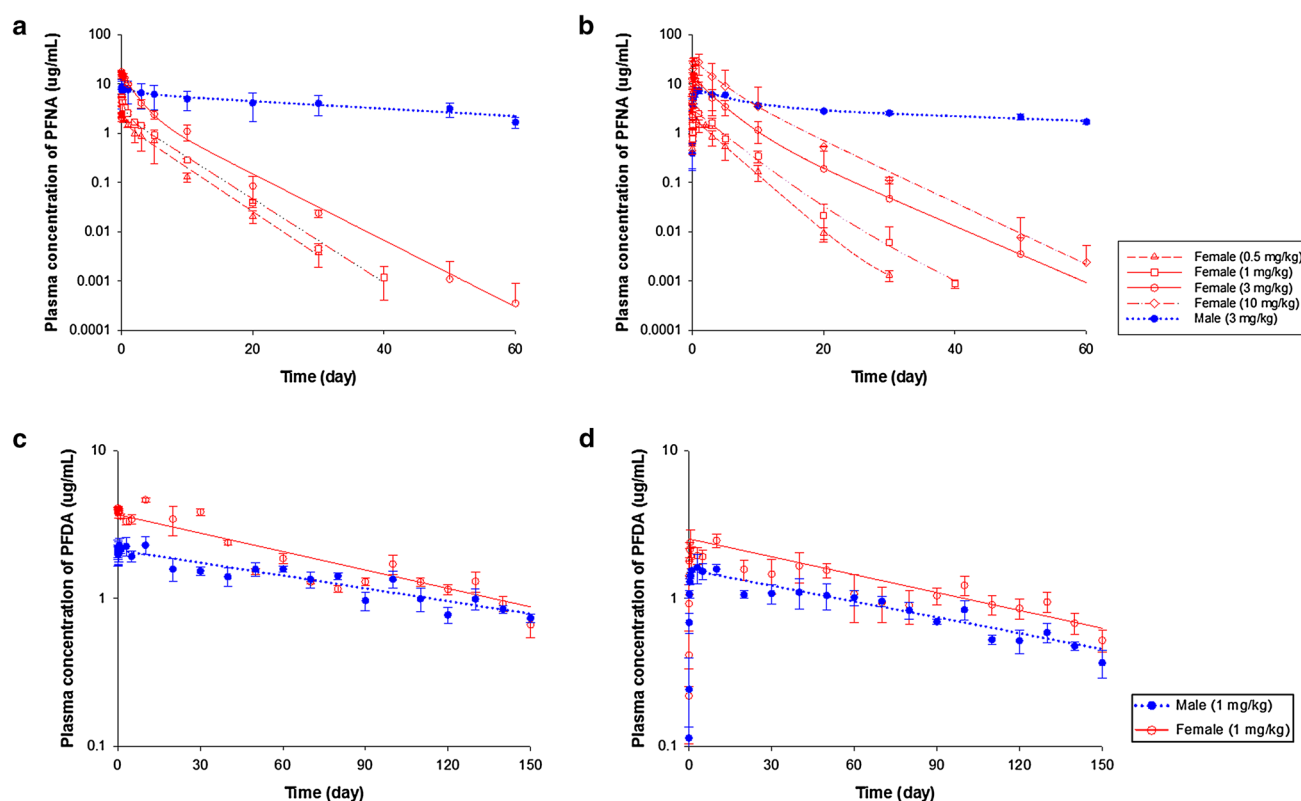


Fig. 3 Mean plasma concentration–time profiles of PFNA (upper panel) and PFDA (lower panel) after IV and oral administration to male and female rats. PFNA was administered to male at a dose of 3 mg/kg and female rats at a dose of 0.5, 1, 3, and (10) mg/kg via **a** IV and **b** oral administration (numbers in parenthesis indicate oral

administration). Male and female rats were given at dose of 1 mg/kg of PFDA through **c** IV and **d** oral administration. The PFNA concentration in female rats given at a dose of 3 mg/kg was measured at LLOQ at 60 days. Each value represented the mean $\pm$ SD ($n=5$)

manner from 8.11 to 45.31 $\mu\text{g}\cdot\text{day}/\text{mL}$ and the $t_{\beta,1/2}$ of 3.38–4.44 days was constant regardless of the administered dose in female rats after IV administration (Fig. 4). After oral administration to female rats, the $C_{\max}$ (1.90–28.38 $\mu\text{g}/\text{mL}$) showed dose dependency. In the male rats after IV administration of PFNA at a dose of 3 mg/kg, the AUC_{inf} , $t_{\beta,1/2}$, and CL were $411.40 \pm 79.67 \mu\text{g}\cdot\text{day}/\text{mL}$, 40.20 ± 18.68 days, and $7.43 \pm 1.44 \text{ mL}/\text{day}$ in male rats, and $45.31 \pm 3.51 \mu\text{g}\cdot\text{day}/\text{mL}$, 4.44 ± 0.16 days, and $16.55 \pm 1.28 \text{ mL}/\text{day}$ in female rats, respectively. The dose relationship of male rats was not estimated to conduct the minimum of animal experiments, but we can expect a dose-dependent of PFNA in male rats based on the results of female rats. The volume of distribution (V_d) was $45.86 \pm 3.65 \text{ mL}$ for female and $363.09 \pm 18.269 \text{ mL}$ for male rats. When comparing the PK parameters of PFNA, the $t_{\beta,1/2}$, CL , and V_d showed significant gender difference in male and female rats. The $t_{\beta,1/2}$ of male rats was about ten times longer than that of female rats, whereas the CL of female rats was approximately 2.5 times higher than that of male rats. The V_d of male rats was about eight times higher than that of female rats. The PK parameters of PFDA are listed in Table 2 and were calculated using

a one-compartment model after IV or oral administration at a dose of 1 mg/kg of PFDA. The mean plasma concentration–time profiles of PFDA are shown in Fig. 3c, d. For the IV administration, the $t_{1/2}$ and CL were 50.03 ± 2.28 days and $0.81 \pm 0.06 \text{ mL}/\text{day}/\text{kg}$ in female rats, respectively, and 109.44 ± 18.72 days and $0.76 \pm 0.10 \text{ mL}/\text{day}$ in male rats, respectively. The V_d was $58.42 \pm 4.46 \text{ mL}$ in female rats and $118.18 \pm 9.31 \text{ mL}$ in male rats. The $t_{1/2}$ and V_d of PFDA showed the gender difference in both male and female rats, whereas CL of male rats was similar to that of female rats. Typically, the PFNA and PFDA were rapidly eliminated in female rats. The PFDA showed a long $t_{1/2}$ and lower CL , compared to the PK parameters of PFNA. Regardless of the sex, the bioavailability of PFNA and PFDA was estimated to be about 0.75–0.79 and 0.65–0.87, respectively; we therefore assumed that the gender difference of $t_{1/2}$ and V_d for PFNA and PFDA is due to the distribution or the excretion of PFNA and PFDA rather than to the absorption.

For the study of tissue distribution, male and female rats were intravenously administered the PFNA (at a dose of the 3 mg/kg) and PFDA (at a dose of 1 mg/kg). Figure 5 shows the K_p values of PFNA and PFDA in male and female rats.

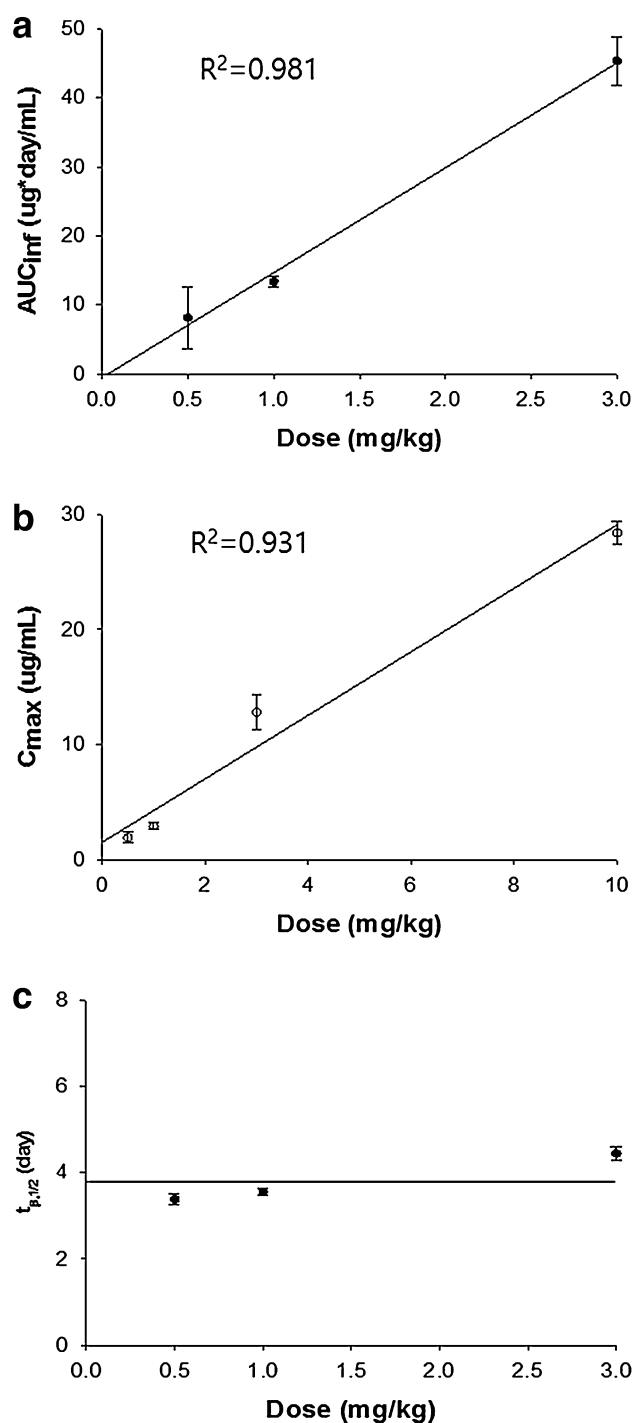


Fig. 4 Dose relationship with **a** AUC_{inf} , **b** C_{max} , and **c** $t_{1/2}$ of PFNA in female rats. The AUC_{inf} and $t_{1/2}$ were obtained from female rats after IV (closed circle) injection at dose of 0.3–3 mg/kg PFNA. The C_{max} was estimated after oral (open circle) administration at dose of 0.5–10 mg/kg PFNA to female rats. Each value represents the mean $\pm$ SD ($n=5$)

The brain, heart, lung, kidney, liver, spleen, GI tract, adipose tissue, and muscle were examined to obtain the K_p values of PFNA and PFDA. The liver showed the highest distribution

of PFNA (1.186, male rats; 0.466, female rats), followed by the kidney (0.128, male rats; 0.247, female rats), and the lung (0.029, male rats; 0.059, female rats). The liver distribution of PFNA in male rats was about 2.5 times higher than that in female rats, whereas the K_p values of kidney in female rats were significantly higher than those in male rats. The tissue distribution of PFNA showed significant gender differences in all tissues except for GI tract and muscles. Consequently, we assumed that the K_p values of PFNA in the liver have a relatively significant effect on the gender differences of $t_{1/2}$ and CL of PFNA. PFDA was also strongly detected in the liver (0.744, male rats; 0.607, female rats), followed by the kidney (0.169, male rats; 0.233, female rats), and lung (0.083, male rats; 0.100, female rats). For all tissues, the mean K_p values of PFDA in male rats were slightly higher than those of female rats, however, the K_p value of the liver only showed significant difference between male and female rats ($p < 0.05$).

The urinary and fecal excretions of PFNA were estimated after IV administration to rats at a dose of 3 mg/kg of PFNA. The mean cumulative excretion percentages of PFNA in urine and feces are shown in Fig. 6. After IV administration, the cumulative excreted percentages of PFNA in urine and feces were $14.33 \pm 9.30\%$ and $1.28 \pm 0.45\%$ in male rats and $34.56 \pm 2.21\%$ and $3.13 \pm 2.18\%$ in female rats, respectively. These results indicate that urine is a major elimination route of PFNA in male and female rats, while higher concentration of PFNA was excreted in female rats than in male rats. The urinary elimination of PFNA showed a gender differences between male and female at each collected time. In the case of PFDA, the cumulative excretion percentages of urine and feces after IV injection were $11.22 \pm 2.96\%$ and $18.25 \pm 2.72\%$ in male rats, respectively, and $22.17 \pm 5.28\%$ and $16.44 \pm 0.70\%$ in female rats, respectively. PFDA was mainly eliminated by feces in male rats, but in female rats, it was more excreted via urine. Fecal elimination of PFDA was slightly higher in female than in male rats. The fecal elimination of PFDA was not significant gender difference in male and female rats, whereas the urinary excretion showed a significant difference.

Protein binding assay in rat and human plasma

The plasma protein binding ratio for PFNA and PFDA was estimated in rats and humans for both male and female. The PFNA and PFDA were strongly bound to plasma protein. The percentage of plasma protein binding of PFNA was calculated as $99.73 \pm 0.04\%$ (male) and $99.67 \pm 0.01\%$ (female) in rats, and 99.85% (male) and 99.88% (female) in humans. The PFDA was bound to the plasma protein by $99.88 \pm 0.005\%$ in male rats, $99.99 \pm 0.002\%$ in female rats, $99.99 \pm 0.002\%$ in male humans, and $99.99 \pm 0.004\%$ in female humans, respectively.

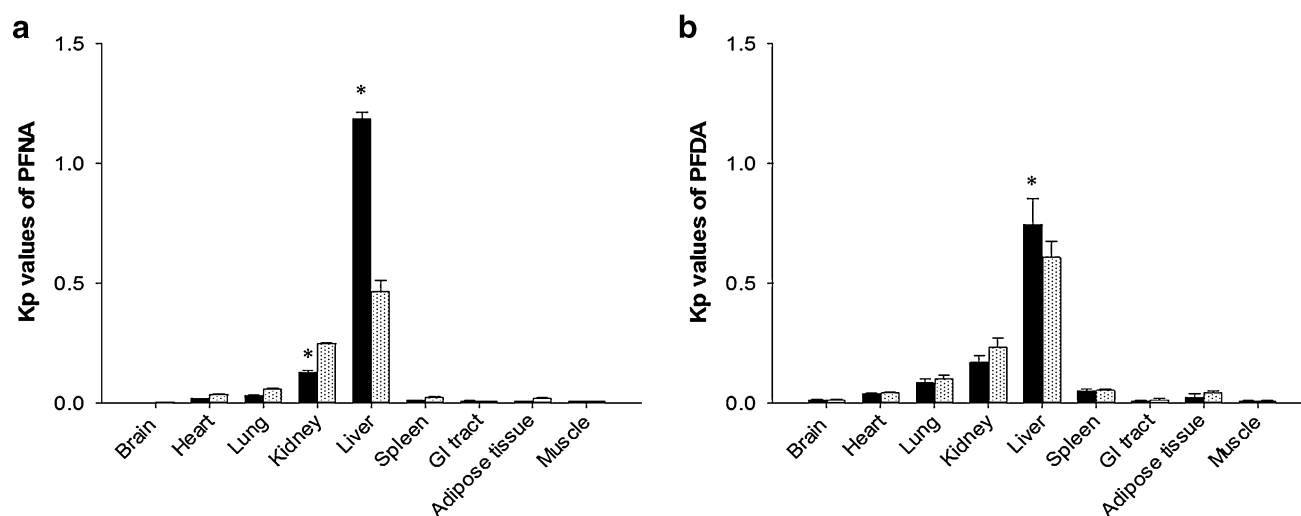


Fig. 5 Partition coefficient (K_p) of **a** PFNA and **b** PFDA in male (black) and female (dotted) rats after IV administration of PFNA (3 mg/kg) and PFDA (1 mg/kg), respectively. Each value represents the mean $\pm$ SD ($n=5$). * $p < 0.05$ between male and female

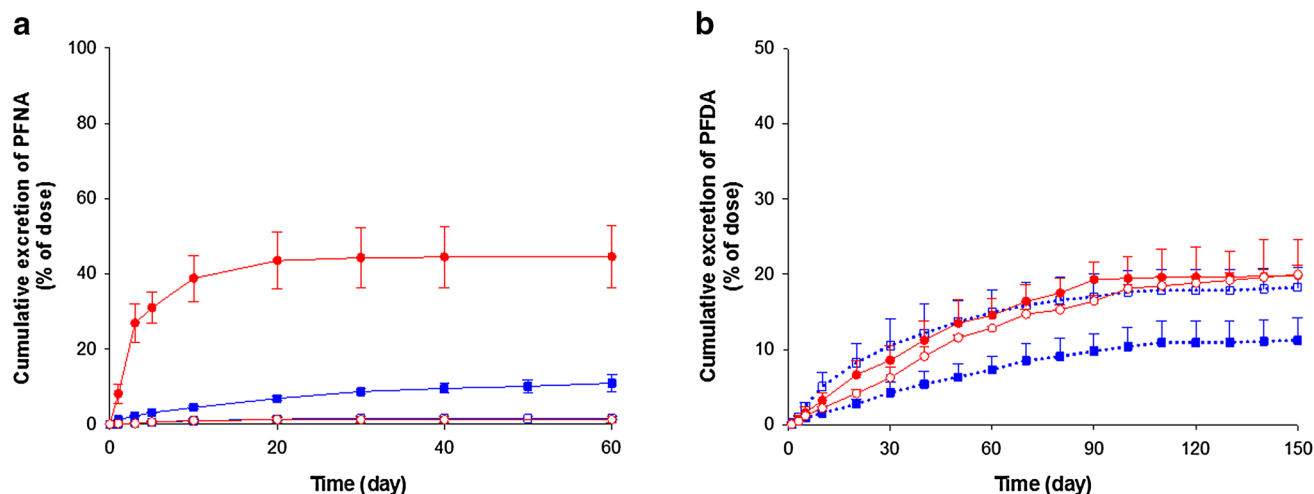


Fig. 6 Cumulative excretion of **a** PFNA and **b** PFDA in rat urine (closed) and feces (open). Male (dotted line, square) and female rats (line, circle) were given at dose of 3 mg/kg (PFNA) and 1 mg/kg (PFDA) via IV injection. The PFNA and PFDA in urine showed a significant difference between male and female rats. Each value represents the mean $\pm$ SD ($n=5$). * $p < 0.05$ between male and female

Development and validation of PBPK model in rats

The PBPK model was established by simulation based on the physiological parameters of rats and experimental data which are obtained from the concentration of each PFAS rat biological matrix after oral administration of PFNA (3 mg/kg) and PFDA (1 mg/kg). The K_p values used were obtained from the rats after IV injection in consistent with dose. Model fitting was performed on the changes of the biochemical parameters, and was visually compared to the experimental values and predicted line. The estimated parameters were summarized in Table 3. For the PFNA, the parameters of T_m , K_t , and K_u were 81,030 $\mu\text{g}/\text{h}/\text{kg}$, 24,850 $\mu\text{g}/\text{L}$, and

3.52 h^{-1} in male rats, respectively, and were 34,530 $\mu\text{g}/\text{h}/\text{kg}$, 24,850 $\mu\text{g}/\text{L}$, and 8.83 h^{-1} in female rats, respectively. The T_m , K_t , and K_u of PFDA were 83,953 $\mu\text{g}/\text{h}/\text{kg}$, 30,051 $\mu\text{g}/\text{L}$, and 0.391 h^{-1} in male rat, respectively, and 40,250 $\mu\text{g}/\text{h}/\text{kg}$, 30,051 $\mu\text{g}/\text{L}$, and 0.681 h^{-1} in female rats, respectively. The predicted and observed concentrations (mean $\pm$ SD) of PFNA in plasma, liver, kidney, and urine after oral administration of PFNA (3 mg/kg) to male and female rats are shown in Fig. 7. Figure 8 shows the predicted and observed concentrations (mean $\pm$ SD) of PFDA in plasma, liver, kidney, and urine after oral administration of PFDA (1 mg/kg) to male and female rats. The fitted concentration of PFNA and PFDA in plasma, tissues, and urine was in good agreement

Table 3 Comparison of gender differences in biochemical parameters of PFNA and PFDA for PBPK modeling

Parameters	Description	PFNA				PFDA				Differences ^c	
		Rat		Human		Rat		Human			
		Male	Female	Male	Female	Male	Female	Male	Female	Male	Female
T_m	Transport maximum ($\mu\text{g/h}$) ^a	81,030	34,530	1,960,963	725,717	83,953	40,250	2,031,701	845,935	>	
K_t	Transport affinity constant ($\mu\text{g/L}$) ^b	24,850	24,850	24,850	24,850	30,051	30,051	30,051	30,051	=	
K_u	Rate constant to urine (h^{-1}) ^b	3.52	8.83	0.861	2.16	0.391	0.681	0.0956	0.174	<	
K_f	Transfer rate constant from GI tract to fecal elimination (h^{-1}) ^b	1.22	0.950	0.298	0.232	0.382	0.391	0.0934	0.100	=	
K_{bile}	Biliary excretion rate to storage (h^{-1}) ^b	–	–	–	–	0.000995	0.00119	0.000243	0.000305	=	
K_{abs}	Oral absorption rate constant (h^{-1}) ^b	6.24	1.20	1.525	0.293	3.84	1.44	0.939	0.369	>	
F_r	Free fraction in blood	0.00272	0.00332	0.00148	0.00122	0.00118	0.000122	0.00157	0.00123	–	
K_{st}	Rate constant to storage compartment (h^{-1}) ^b	0.371	2.01	0.0910	0.491	0.148	0.758	0.0362	0.194	<	

^aThe T_m in humans was scaled as $\text{BW}^{0.75}$ ^b k_{human} means the rate constant in humans and is calculated from Eq. (4)^cDifferences illustrate a significant difference of biochemical parameters between male (M) and female (F)

with the measured concentration in male and female rats. The GI tract as an absorption site and filtrate compartment was included in the PBPK model for PFNA and PFDA to explain the renal reabsorption. In case of PFDA, the storage compartment was added additionally to describe the fecal elimination regarding the major excretion routes of PFDA by the previous reports (Kudo et al, 2001; Loccisano et al. 2012a). The brain, muscle, spleen, and adipose tissue were classified as belonging to the rest of body due to the low level of PFNA and PFDA in these tissues.

Figure 9 shows the normalized sensitivity coefficients of PFNA and PFDA in rats. The model parameters of PFNA and PFDA showed a significant gender difference. PFNA was potentially dependent on the F_r (–1.02) and K_f (–0.7) in male rats, and the F_r (–1.30) and Q_{fil} (–1.13) for female rats. In addition, the parameters of T_m (0.31), K_t (–0.30), Q_{fil} (–0.33), and K_{st} (–0.35) have a similar effect in male rats. The T_m (0.88), K_t (–0.76), and K_{st} (–0.76) were similar in female rats. These parameters showed more dependence in female rats than that of male rats. The other parameters were less than 0.2. PFDA was highly influenced by F_r (–0.83, male; –0.79, female) and Q_{fil} (–0.52, male; –0.85, female) in rats. In the male rats, T_m (0.51), K_t (–0.50), Q_{fil} (–0.52), and K_{st} (–0.50) showed the similar influence, and then K_f and K_{bile} were –0.32 and –0.18, respectively. PFDA in female rats was depended on the parameter of T_m (0.48), K_t (–0.46), and K_{st} (–0.46), respectively. The other parameters were less than 0.1 in male (K_u and K_{abs}) and female (K_u , K_f , K_{bile} , and K_{abs}) rats.

Extrapolation to the PBPK model in humans

The structure of the PBPK model of PFNA and PFDA in humans is the same that in rats (Fig. 2). The predicted concentration–time profiles of PFNA and PFDA in plasma, tissues, and urine of male and female humans are shown in Figs. 10 and 11. In humans, the AUC_{inf} of each PFAS was calculated from the predicted plasma concentration in humans. The AUC_{inf} of PFNA was 626.5 $\mu\text{g}\cdot\text{day}/\text{mL}$ and 163.2 $\mu\text{g}\cdot\text{day}/\text{mL}$ in the male and female humans, respectively, after the oral administration of 1 mg/kg. The AUC_{inf} of PFDA in male and female humans was 2336.5 $\mu\text{g}\cdot\text{day}/\text{mL}$ and 1674.2 $\mu\text{g}\cdot\text{day}/\text{mL}$ after the oral administration of PFDA at a dose of 1 mg/kg to male and female humans. The AUC_{inf} of PFNA and PFDA in male rats was approximately four (PFNA) and two (PFDA) times higher than male rats. The liver and kidney have a higher level of two PFASs than other tissues.

Human health risk assessment

The human health risk assessment was carried out using the rat NOAEL with 0.2 mg/kg/day with hepatic toxicology

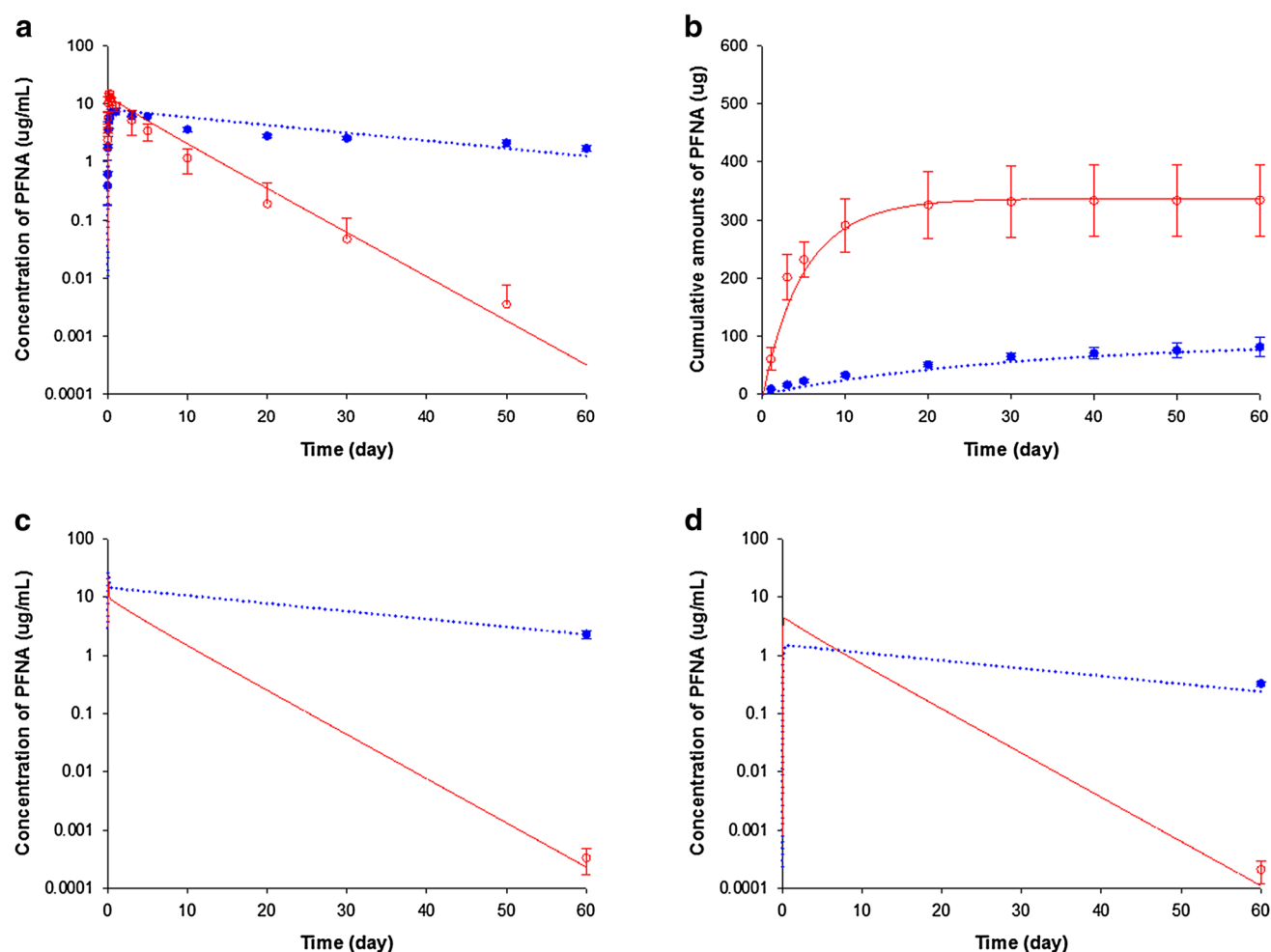


Fig. 7 Fitted (line) and observed (circle) concentration of PFNA in rat **a** plasma, **b** urine, **c** liver, **d** kidney after oral administration at doses of 3 mg/kg PFNA in male (dotted line) and female (line) rats. Open circles refer to the female rats and closed circles are male

rats. The fitted line of PFNA obtained from a rat PBPK model was in good agreement with observed values. Each value represents the mean $\pm$ SD ($n=5$)

for PFNA (Fang et al. 2012b) and 1.2 mg/kg/day with toxic hepatopathy for PFDA (Kawashima et al. 1995) as the most sensitive POD. For the PFNA, the PODs in human were calculated at 8.18 $\mu\text{g/kg/day}$ (male) and 8.89 $\mu\text{g/kg/day}$ (female), according to the dose-metric approach. The TDIs were 81.8 ng/kg/day in males and 0.889 $\mu\text{g/kg/day}$ in females, applying a UF_H (10) and UF_S (10) for human POD. External doses of 3.35 ng/kg/day (male) and 17.0 ng/kg/day (female) in humans were estimated based on the Korean biomonitoring values (male, 2.30 ng/mL; female, 2.90 ng/mL). The MOEs of PFNA were estimated to be 23.2 in male and 5.24 in female humans, respectively. The results suggest that the MOE of human males was about four times higher than that of females. The PFDA of risk assessment was estimated in the same manner as PFNA. In the case of PFDA, the PODs in humans were estimated at 28.4 $\mu\text{g/kg/day}$ and 67.6 $\mu\text{g/kg/day}$ in male and female, respectively, and the

TDIs in male and female humans were 2.48 $\mu\text{g/kg/day}$ and 6.67 $\mu\text{g/kg/day}$, respectively. External doses were calculated at 0.530 ng/kg/day in male and 0.661 ng/kg/day in female humans, based on the Korean biomonitoring values (male 1.09 ng/mL, female 1.01 ng/mL). The MOEs for PFDA were estimated to be 536 in male and 1022 in female humans, respectively. MOE of PFDA in human males was about two times higher than that of females.

Discussion

In the previous publications, Tatum-Gibbs et al. (2011) reported the PK parameters of PFNA in male and female rats after single oral administration at a dose of 1–10 mg/kg of PFNA (Tatum-Gibbs et al. 2011). While the $t_{\alpha,1/2}$ of female rats (1.02–2.08 days) in that study was similar to that

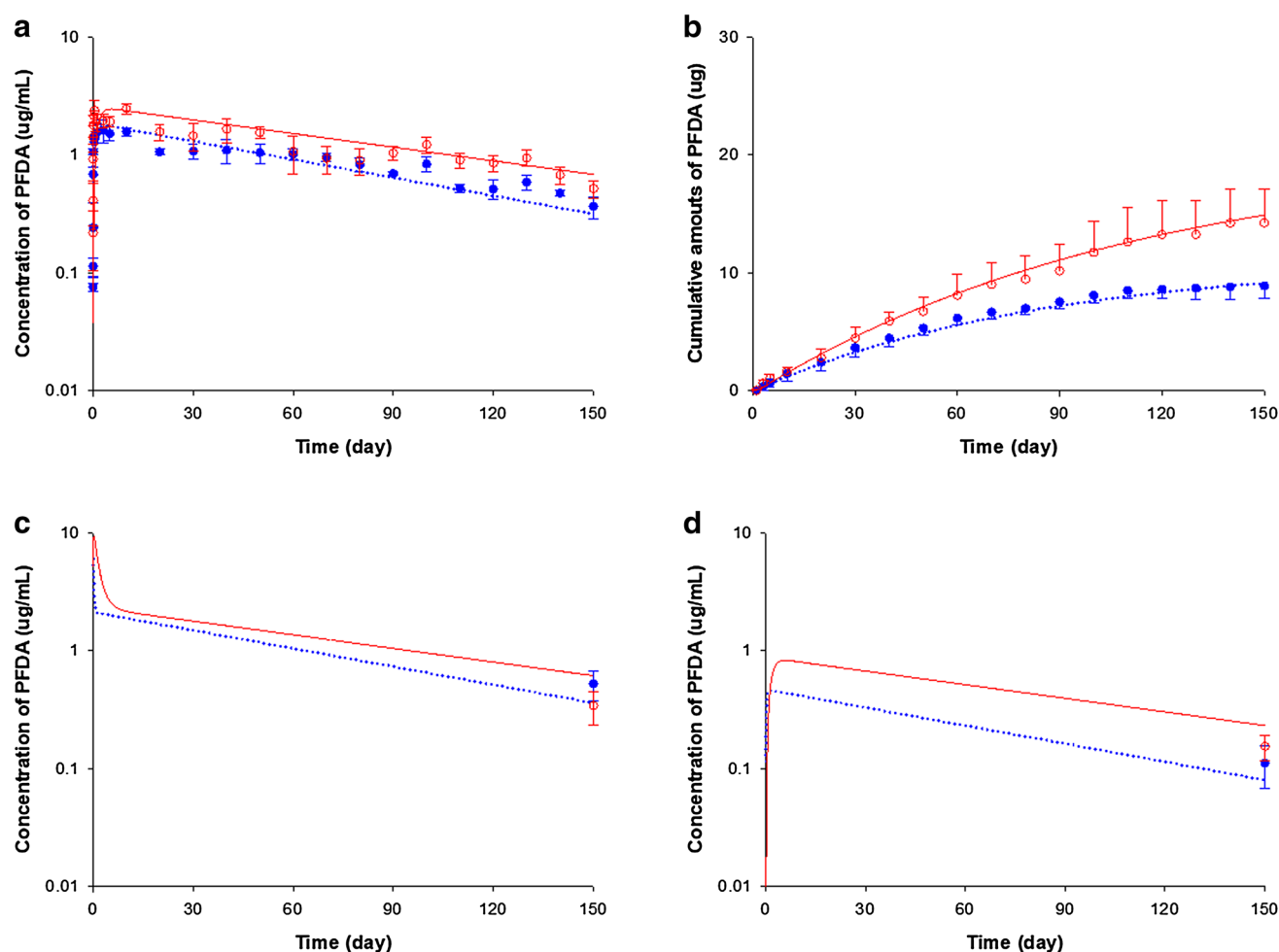


Fig. 8 Fitted (line) and observed (circle) concentration of PFDA in rat **a** plasma, **b** urine, **c** liver, **d** kidney after oral administration at doses of 1 mg/kg PFDA in male (dotted line) and female (line) rats.

Open circles indicate the female rats and closed circles refer to the male rats. The fitted line and observed values of PFDA by PBPK modeling was well fitted. Each value represents the mean $\pm$ SD ($n=5$)

in our results (1.64–2.72 days), we were not able to compare the results of $t_{\beta,1/2}$ because their data were measured under the background concentration after 16 h. The $t_{\beta,1/2}$ of male rats were ranged from 20.2 to 55.6 days, which corresponded to that of our results (40.20–54.57 days). They suggested that the serum elimination of PFNA showed a gender difference in rats (Tatum-Gibbs et al. 2011). However, CL was not shown in their results and was thus unable to be compared. Ohmori et al. (2003) reported that PFNA was more rapidly eliminated in female rats ($t_{1/2}$, 2.44 days) than in male rats ($t_{1/2}$, 29.57 days) after IV administration of PFNA (48.64 $\mu\text{mol/kg}$) (Ohmori et al. 2003). Similar to the above-mentioned studies, Benskin et al. (2009) reported that the $t_{1/2}$ of PFNA in male rats was 40.6 days (Benskin et al. 2009). In the case of PFDA, Ohmori et al. (2003) reported that the elimination at $t_{1/2}$ of PFDA was 39.92 ± 8.62 days in male rats and 58.57 ± 5.84 days in female rats after IV injection at a dose of 48.64 $\mu\text{mol/kg}$, and showed gender

difference. However, the PK parameters of these results are insufficient to describe the plasma–time concentration profiles. Vanden Heuvel et al. (1991) reported that the $t_{1/2s}$ of PFDA in female rats (29 days) was slightly longer than that in male rats (22 days), but they did not show a gender difference. In contrast, Kudo et al. (2000) reported that the serum concentration of PFDA was significantly ($p < 0.05$) different in male and female rats (PFDA serum level of female rats was higher than that of male rats) and their results were consistent with our results. In the present study, PFDA showed a higher plasma concentration in female rats than in male rats, whereas it was rapidly eliminated in female rats (The $t_{1/2}$ of PFDA was 109.44 ± 18.72 days in male rats and 50.03 ± 2.28 days.). We expected this was due to the difference in the biliary excretion between male and female rats. Kudo et al. (2000) reported that a strong induction of peroxisomal β -oxidation in the liver indicates a higher accumulation of PFASs. The PFDA was induced a strong peroxisomal

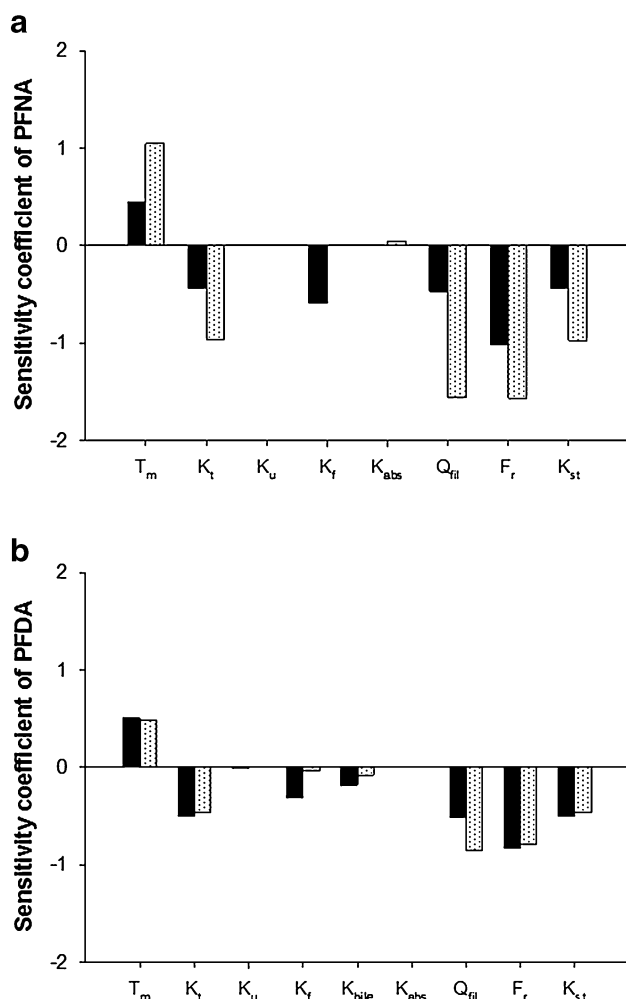


Fig. 9 Sensitivity coefficients for PBPK model parameters for **a** PFNA and **b** PFDA in male (black) and female (dotted) rats

β -oxidation and was highly detected in the liver in both male and female rats, although the biliary excretion rate of PFDA in female rats was higher than that of male rats (Kudo et al. 2001). However, while the urinary and fecal elimination of PFDA did not show significant gender difference, the fecal elimination of female rats was slightly higher than that of male rats. While the reason for this has not yet been clearly identified, but it may be caused by enterohepatic circulation. In a long-term experiment, Kudo et al. (2001) suggested that enterohepatic circulation could be responsible for the elimination of PFASs and could be ascribed to the rapid clearance of PFOA (Kudo et al. 2001). In addition, we performed the *in vivo* study for PFNA at dose of 0.5, 1, and 3 mg/kg (for IV injection) and 0.5, 1, 3, and 10 mg/kg (for oral administration) to evaluate the dose-dependency for PFNA. In case of oral administration, we designed the experiment by adding a dose of 10 mg/kg PFNA, considering the limit of quantitation in biological samples (organ and tissues, urine, and feces) and bioavailability of the

PFNA. We confirmed that PFNA showed a dose-dependency in ranges of 0.5–3 mg/kg dose (IV administration) and 0.5–10 mg/kg dose (oral administration). In our previous study, the animal study design to identify the gender differences of PFOA, PFOS, and PFHxS in rats was performed with one dose (Kim et al. 2016b). It was sufficient to compare the gender differences for PFASs with similar structures. For this reason, we planned the animal study design with one dose administration to identify the gender differences in the present study. Therefore, the male rats were received at one dose (3 mg/kg of PFNA, 1 mg/kg of PFDA). PFDA belongs to the PFASs such as PFNA, PFHxS, PFOA, and PFOS, and they have a similar structure and properties with high stability and bioaccumulation. We conducted the *in vivo* study for PFDA considering the PKs of PFNA (in the present study) and PFHxS (Kim et al. 2018), and other PFASs (Kim et al. 2016a, b) which have continuously been studied. We expected to show the dose-dependence of PFDA might be similar to that of PFNA and PFHxS. The AUC_{inf} was increased in dose-dependence for PFNA (Fig. 4) at dose ranged from 0.5 to 3 mg/kg and PFHxS (Kim et al. 2018) at dose ranged from 0.5 to 10 mg/mL. The $t_{1/2}$ was independent to dose in both of PFNA and PFHxS. Therefore, we examined at one dose (1 mg/kg) for PFDA.

For the tissue distribution of PFASs, the PFNA showed the greatest accumulation in rat liver (Tatum-Gibbs et al. 2011). At 50 days after PFNA exposure, the highest concentration of PFNA in male rats was detected in the liver, whereas that in female rats was estimated to be below the lower limit of quantification (LLOQ) (Tatum-Gibbs et al. 2011). In the liver, PFNA was more rapidly eliminated in female rats than in male rats. These results are consistent with our results. Guruge et al. (2016) reported on the tissue distribution of nine PFASs at 21 days after exposure of PFASs to female microminipigs (Guruge et al. 2016). PFNA and PFDA in microminipigs were highly detected in the liver and kidney. The PFDA was also highly distributed in the liver and kidney (Benskin et al. 2009; Vanden Heuvel et al. 1991). Kudo et al. (2001) showed that the liver level of PFNA and PFDA was a significantly different in male and female rats (Kudo et al. 2001). In the present study, we reported that PFNA and PFDA were highly distributed in the liver and kidney and showed a gender differences in liver. These results are consistent with those in the above-mentioned studies.

Kudo et al. (2001) reported that PFNA and PFDA were eliminated through the urine at 2% and 0.2% of the dose in male rats, and 51% and 0.2% of the dose in female rats. The fecal elimination of PFNA and PFDA was less than 5% of the dose for 120 h after administration. PFNA showed gender difference in urinary elimination, whereas for PFDA, a difference in male and female rats was not observed and it was mainly excreted via the feces (Kudo et al. 2001). In

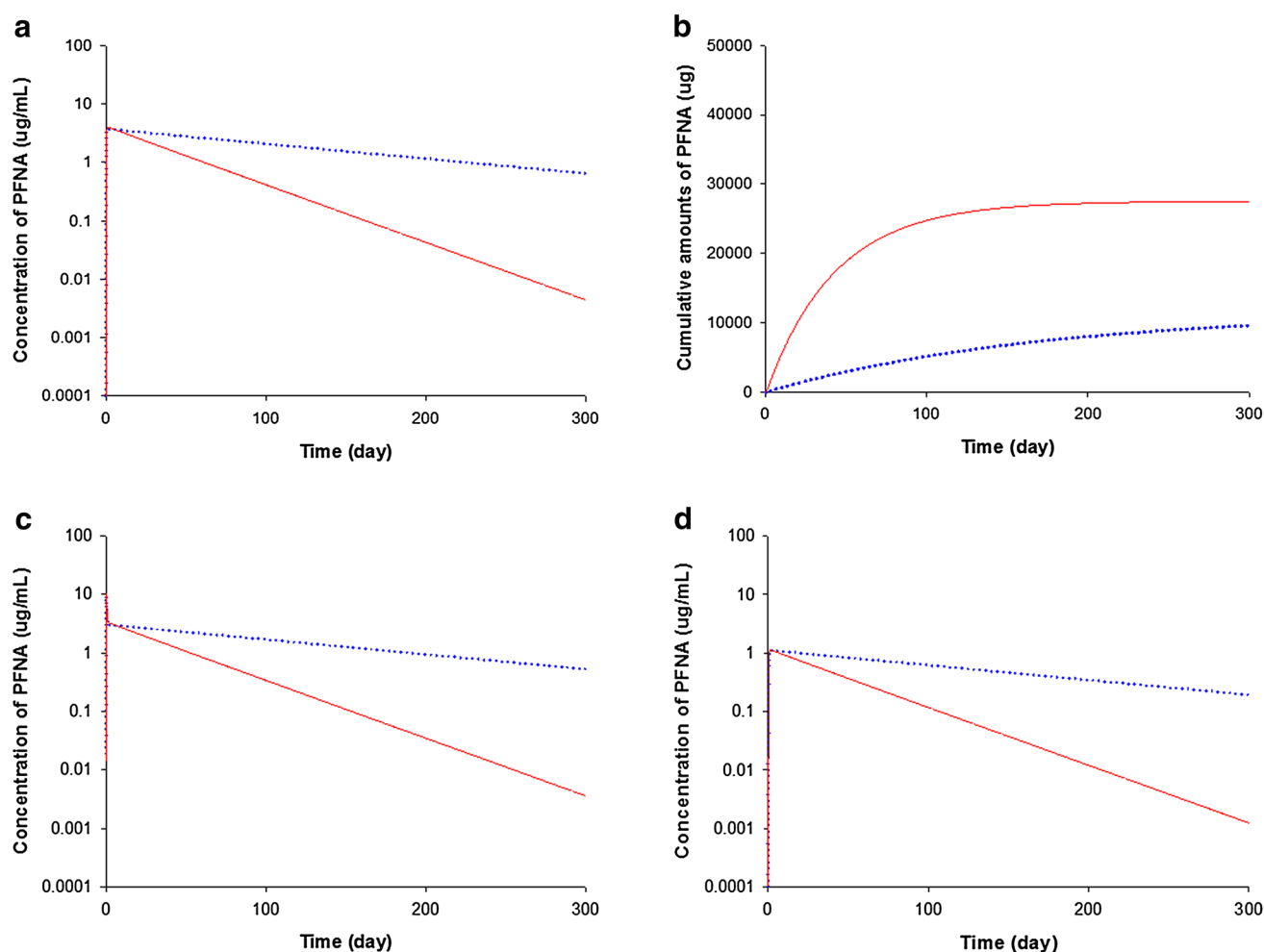


Fig. 10 Simulated concentrations of PFNA in human **a** plasma, **b** urine, **c** liver, and **d** kidney after oral administration at doses of 1 mg/kg PFNA in male (dotted line) and female (line) humans. The con-

centration of PFNA was predicted by the human PBPK model extrapolated from rat PBPK model

other study, PFDA was excreted daily at 0.5% of the dose in male and female rats after intraperitoneal administration of 20 mg/kg through the feces (Ylinen and Auriola 1990). Our results showed that PFNA was mainly eliminated via urine (34.6% of dose, female; 14% of dose, male rats) and showed a significant gender difference. PFDA highly excreted via feces in male rats and the results corresponded to the reports by Kudo et al. (2001). The PFDA in female rats was slightly higher eliminated through the urine although a major elimination route of PFDA was known to be feces (Kudo et al. 2001; Vanden Heuvel et al. 1991). But the urinary and fecal elimination of PFDA in female rats did not show a significant difference. The fecal elimination of PFDA did not appear the significant difference in male and females, whereas the urinary excretion of PFDA showed a gender difference. In addition, we compared the biochemical parameters of PFNA and PFDA, which are obtained from the PBPK modeling. We confirmed that the urinary elimination

of PFDA was higher in female rats (0.681, K_u ; 0.758, K_{st}) than in male (0.391, K_u ; 0.148, K_{st}) rats. The fecal excretion of PFDA showed no gender difference between male and female rats based on the K_f (0.382, male; 0.391, female) and K_{bile} (0.000995, male; 0.00119, female) parameters.

The PK parameters of PFOA, PFOS, and PFHxS investigated previously were compared. The length of $t_{1/2}$ in female rats was long in the order of PFDA (50.03 days), PFOS (24.80 days), PFNA (3.79 days), PFHxS (0.88 days), and PFOA (0.19 days), while in case of male rats, the length of $t_{1/2}$ was long in the order of PFDA (194.05 days), PFNA (40.20 days), PFOS (28.7 days), PFHxS (20.7 days), and PFOA (1.64 days) (Kim et al. 2016b). The results show that the PFASs had a longer carbon chain length indicated by the longer $t_{1/2}$ in male rats, except for PFOS and PFHxS contained a $-SO_3H$. The numbers of carbons for PFASs were PFDA (C10) > PFNA (C9) > PFOS (C8) > PFHxS (C6) > PFOA (C8).

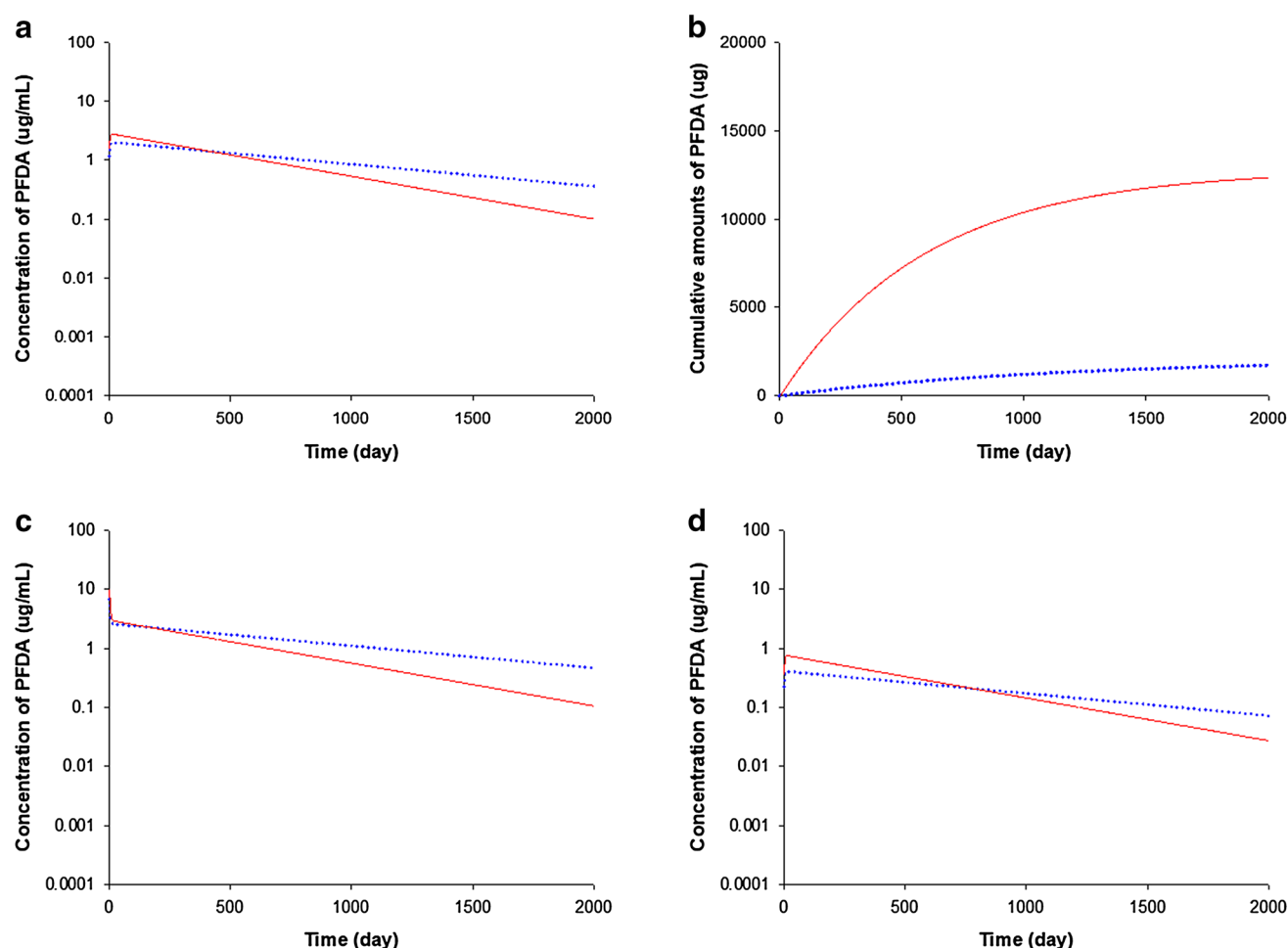


Fig. 11 Simulated concentrations of PFDA in human **a** plasma, **b** urine, **c** liver, and **d** kidney after oral administration at doses of 1 mg/kg PFDA in male (dotted line) and female (line) humans. The predicted concentration of PFDA was obtained from the simulation of human PBPK model

For the distribution of PFASs, PFOA, PFOS, and PFHxS were highly cumulative in the liver and kidney (Kim et al. 2016a, b). The highest levels of PFASs were detected in the liver, and the PFOA (male), PFOS (male and female), and PFNA (male) showed K_p values of more than 1. These results could be influenced by the PFASs and by the liver fatty acids binding protein (Luebker et al. 2002) and activity of peroxisomal β -oxidation in the liver (Kudo et al. 2000). The PFASs with a shorter carbon chain length are rapidly eliminated in rats. After intraperitoneal administration to male rats, perfluoroheptanoic acid (PFHA), PFOA, PFNA, and PFDA were eliminated in the urine at levels of 92%, 55%, 2.0%, and 0.2% within 120 h, respectively. In the female rats, PFOA and PFNA were more rapidly eliminated via urine with 80% and 51% eliminated within 120 h, respectively (Kudo et al. 2001). PFOA and PFHxS were more rapidly excreted in female rats (52.8% of dose, PFOA; 28.0% of dose, PFHxS) than in male rats (9.2% of dose, PFOA; 8.3% of dose, PFHxS) after IV injection, whereas the

urinary elimination of PFOS was similar in male (20.6% of dose) and female rats (17.7% of dose) (Kim et al. 2016b). In the present study, PFNA and PFDA, which have a relatively long carbon length, were more slowly excreted through the urine in male rats (14.3%, PFNA; 11.2%, PFDA) than in female rats (34.7%, PFNA; 19.3%, PFDA). The order of elimination through urine was PFOA (52.8%) > PFHxS (28.0%) > PFOS (17.7%), PFNA (14.4%) > PFDA (11.2%) in female rats (Kim et al. 2016b). From these results, the PFASs with shorter carbon chain length are rapidly excreted in via urine and appeared to have a short $t_{1/2}$. PFDA was reported that a major elimination route was feces (Kudo et al. 2001; Vanden Heuvel et al. 1991). PFDA did not show significant difference between urine and fecal elimination in female rats, although urinary elimination of female rats was higher than that of male rats.

The PFASs were known to reabsorbed kidney by renal organic anion transport (OAT) (Han et al. 2012). Han et al. (2012) reported the differences in the renal OAT among

human, rat, and mouse kidneys. The PFASs are excreted by Oat1 and Oat3, and are reabsorbed by Oatp1a1 (Weaver et al. 2010). This mechanism was appropriately applied to PFOA (Loccisano et al. 2013) and it was shown that other PFASs which might be responsible for Oat1, Oat3, and Oatp1a1. The mechanism of PFHxS and PFNA for renal OAT was not clearly reported. Moreover, while PFOA and PFHxS in female rats with short $t_{1/2}$ were rapidly eliminated via urine, in contrary, in male rats with long $t_{1/2}$, they are slowly eliminated (Kim et al. 2018). The gender difference of PFNA and PFDA, the structure of which is similar to PFOA and PFHxS, is predicted for the same reason.

PFASs are known to bind to plasma albumin in humans (Han et al. 2003). PFNA was highly bound to bovine serum albumin (BSA) or human serum albumin (HSA) by more than 99.9% (Bischel et al. 2010). 99% of PFDA was bound by the serum proteins (Ylinen and Auriola 1990). We obtained results consistent with previous reports. In the present study, PFNA and PFDA were bounded to plasma protein by more than 99.7% in rat and human plasma. Similarly, Vanden Heuvel et al. (1991) reported that PFDA and PFOA covalently bind proteins when administered to rats.

In the sensitivity analysis, the parameters of T_m , K_r , Q_{fil} , F_r , K_f and K_{st} strongly influenced by PBPK modeling of PFNA and PFDA in male and female rats. In the PFNA, the values of T_m (0.31, male; 0.88, female), K_t (−0.30, male; −0.76, female), Q_{fil} (−0.33, male; −1.13, female), K_f (−0.70, male; −0.18, female), and K_{st} (−0.35, male; −0.76, female) in female rats were at least two times higher than that of male rats, whereas the F_r was similar both gender (−1.02, male; −1.30, female). These results indicate that female rats were strongly influenced by T_m and K_t which are a key parameter for reabsorption to kidney. Like the PFNA, PFDA also depended the parameters of T_m (0.51, male; 0.48, female), K_t (−0.50, male; −0.46, female), Q_{fil} (−0.52, male; −0.85, female), and K_{st} (−0.49, male; −0.46, female), but there are no sex differences. However, the parameters of K_f (−0.32, male; −0.036, female), and K_{bile} (−0.18, male; −0.081, female), were more sensitivity in male rats. Comparing the other PFASs such as PFOA and PFOS, they also highly depended on the T_m , K_t , K_{st} , F_r . The PFHxS with long $t_{1/2}$ in male rats appeared strong influence, whereas in female rats did not showed the sensitivity (Kim et al. 2018). These results are consistent with the literature by Loccisano et al. (2012a) who suggested the PFOA was dependent on the T_m and K_t in male rats, not female rats (Loccisano et al. 2012a).

PBPK models for PFASs have been reported on PFOA, PFOS, and PFHxS. The PBPK model for PFOA and PFOS was developed in monkeys (Loccisano et al. 2011) and in adult male and female rats (Loccisano et al. 2012a). Their PBPK models suggested a simple structure composed of plasma, liver, kidney, and filtrate compartments. Some studies reported the menstruation as a specific elimination

routes in women (Wong et al. 2014) and describe the association between serum PFASs levels and the early onset of menopause (Ruark et al. 2017) and delayed menarche (Wu et al. 2015). We recently reported on the PBPK model of PFHxS for human risk assessment (Kim et al. 2018). The PBPK model was composed of plasma, filtrate, and storage compartment, tissues (brain, heart, lung, kidney, liver, GI tract) and the rest of body. Moreover, the PBPK model of PFNA and PFDA has not yet been published. We adapted and modified the model structure of our previous report, to develop an adequate PBPK model of PFNA and PFDA. The developed PBPK model for PFNA and PFDA included the filtrate and storage compartment, considering the reabsorption to kidney of PFASs. Regarding the major elimination route of PFDA in the literatures (Kudo et al. 2001; Loccisano et al. 2012a), the model of PFDA was considered on the biliary excretion and added in storage compartment from liver to feces. The biliary excretion rates of PFDA were obtained from the fitting between predicted line and observed concentration of plasma and feces. The predicted and measured concentration of PFDA was fitted well. The value of biliary excretion rates was estimated at 0.000995 (male) and 0.00119 (female) in this model, and there are no significant gender differences. These results consisted with the literature by Vanden Heuvel et al. (1991) who reported that gender difference in biliary excretion was not appeared. Moreover, it means that the fecal elimination was not different in male and female rats. The developed PBPK model for PFNA and PFDA was included the menstrual blood loss in female human (Kim et al. 2018), while the brain, muscle, spleen, and adipose tissues were classified as the rest of body in the model, considering the remaining body tissues. We examined the PBPK model of PFNA and PFDA with or without the menstrual blood loss in female humans. The effect on menstruation was not observed in the model simulation. It would be due to small volume of menstrual blood (1.61% of total blood volume) (Kim et al. 2018). Therefore, the menstruation would be no influence on the exposure of PFASs according to the gender differences. The PBPK models in this study explain the kinetics of PFNA and PFDA in the body based on the gender differences in detail.

In the present study, some biochemical parameters of PFNA and PFDA showed gender difference among male and female rats. The T_m values for PFNA were 81,030 $\mu\text{g}/\text{h}$ (male) and 34,530 $\mu\text{g}/\text{h}$ (female) and those for PFDA were 83,953 (male) and 40,250 $\mu\text{g}/\text{h}$ (female). These results were consistent with those for PFASs in other studies that showed a similar tendency for gender differences as those for PFOA (95,459 $\mu\text{g}/\text{h}$, male; 1061 $\mu\text{g}/\text{h}$, female) and PFHxS (85,300 $\mu\text{g}/\text{h}$, male; 910 $\mu\text{g}/\text{h}$, female) (Kim et al. 2018; Loccisano et al. 2012a). The T_m value was higher in male rats than in female rats. The T_m of PFOS, which did not show gender difference, was the same in male and

female rats (Loccisano et al. 2012a). In addition, PFASs that have a high value of K_t showed a long $t_{1/2}$. In the present study, the K_t of PFNA and PFDA was estimated as 24,850 $\mu\text{g/L}$ and 30,051 $\mu\text{g/L}$, respectively, in both male and female rats. Comparing the K_t of PFOA, PFOS (Loccisano et al. 2012a), and PFHxS (Kim et al. 2018) in previous studies, the K_t was higher in the present study in order of PFDA (30,051 $\mu\text{g/L}$) > PFNA (24,850 $\mu\text{g/L}$) > PFOS (16,700 $\mu\text{g/L}$) > PFHxS (15,851 $\mu\text{g/L}$) > PFOA (696.8 $\mu\text{g/L}$). The $t_{1/2}$ in rats was reported as 1.64 days (male) and 0.19 days (female) for PFOA, and 20.7 days (male) and 0.88 days (female) for PFHxS, and 26.4 days (male and female) for PFOS, respectively (Kim et al. 2016a, b). We evaluated the $t_{1/2}$ of PFNA (40.2 days, male; 4.44 days, female) and PFDA (109.44 days, male; 50.03 days, female) in the present study. Overall, when comparing biochemical parameters, it was confirmed that gender differences showed a same tendency for the biochemical parameters of PFNA and PFDA. This may be due to the similar physical properties and chemical structure such as the carbon length of the two perfluorinated compounds.

The free fraction in rats was 0.00272 (male) and 0.00332 (female) for PFNA, and 0.00118 (male) and 0.000122 (female) for PFDA. The free fraction of PFNA in female rats was slightly higher than that of male rats, and PFDA showed a significant gender difference in male and female rats (approximately ten times higher in male rats). In humans, the free fraction was 0.00148 (male) and 0.00122 (female) for PFNA and 0.00157 (male) and 0.00123 (female) for PFDA. Gender differences of free fraction were not observed in male and female humans. Kim et al. (2018) reported that the free fraction of chemicals is responsible for reabsorption in the filtrate compartment and elimination from the filtrate compartment to outside the body; however, it was insufficient for describing the gender differences for PFASs.

While the risk assessment of PFASs has continued to be reported, most studies have focused on PFOA and PFOS, although more than ten PFASs have been tested. Borg et al. (2013) reported on the HQs of PFNA and PFDA in the Swedish population, and indicated no concern for the hepatotoxicity (HQ, 0.014) and reproductive toxicity (HQ, 0.0073) for PFNA and hepatotoxicity (HQ, 0.010) and reproductive toxicity (HQ, 0.005) for PFDA. The HQs of PFNA and PFDA were less than those of PFHxS (<0.0035), PFOA (<0.0083), and PFOS (0.14) for reproductive toxicity (safe, HQ < 1; unsafe, 1 < HQ) (Borg et al. 2013). Although Liu et al. (2015) examined the risk assessment of 11 PFASs in the atmosphere of Shenzhen, China, they only reported on the level of PFOA (2.4E–7) and PFOS (HQ, 3.5E–6), which are abundantly found in these areas (Liu et al. 2015). We recently reported on the risk assessment of PFHxS in Korean and suggested the MOE (100.5, male; 27.7, female) based on the PBPK modeling (Kim et al. 2018). As a further study,

we investigated the risk assessment of PFNA and PFDA in Korean men and women in the same manner. We suggested that the TDI was 81.8 ng/kg/day (male) and 88.8 ng/kg/day (female) for PFNA and 284 ng/kg/day (male) and 676 ng/kg/day (female) for PFDA. Comparing the other PFASs, the TDI of PFNA was similar with PFOA (20–150 ng/kg/day) and higher than PFOS (20–50 ng/kg/day). And TDI of PFDA was similar to PFHxS (711 ng/kg/day, male; 159 ng/kg/day, female) (Dong et al. 2017; Kim et al. 2018). MOEs of PFNA and PFDA were estimated as 23.2 (male) and 5.24 (female) for PFNA and 536 (male) and 1022 (female) for PFDA in humans. The high values of MOEs for PFNA and PFDA indicate no concern in the health risk assessment. So far, the risk assessment for PFASs has focused on PFOA and PFOS. The present study is subsequent to our previous recent study reporting the risk assessment for PFHxS using the PBPK model considering the gender difference. Studies on the PFNA and PFDA in risk assessment are somewhat limited, and few have been reported to date. Therefore, we first reported PFNA and PFDA in risk assessment using PBPK model considering gender difference.

Conclusion

In conclusion, the PBPK model for PFNA and PFDA was successfully developed and validated based on a PK study and the tissue distribution after oral or IV administration to rats. The developed PBPK model in male and female rats was used for extrapolation to the human PBPK model. The PBPK model for PFNA and PFDA was shown to be a useful approach for human health risk assessment in Korean populations regarding the potential for accumulated health problems in humans exposed to PFASs. We found that the serum level of PFNA and PFDA in Korean men and women is not a concern at present. In further study, we intend to construct a PBPK core model covering all of the PFASs related to the human health risk assessment of PFASs.

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Compliance with ethical standards

Conflict of interest The authors have declared no conflict of interest.

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